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Title: Implementation of Collection Interface Methods in Java

1. Objective / Aim

To implement and demonstrate commonly used methods of the Java Collection interface using an ArrayList, including add(), addAll(), contains(), containsAll(), size(), isEmpty(), remove(), removeAll(), iterator(), and clear().

2. System Requirements

Hardware Requirements

Hardware Requirement	Specification
Processor	Intel Core i3 or equivalent and above
RAM	Minimum 4 GB
Storage	Minimum 1 GB free space
Input Device	Keyboard and mouse
Display	Standard monitor

Software Requirements

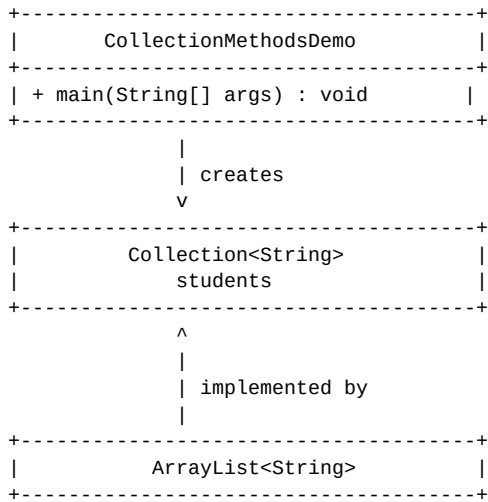
Software Requirement	Specification
Operating System	Windows / Linux / macOS
Java	JDK 17 or JDK 21
IDE / Editor	Eclipse / IntelliJ IDEA / NetBeans / VS Code
Compiler	javac
Execution Environment	Java Virtual Machine (JVM)

3. Algorithm / Logic Description

1. Start the program.
2. Import ArrayList, Arrays, Collection, and Iterator classes.
3. Create a Collection named students using ArrayList.
4. Add initial student names using add().
5. Create another collection using Arrays.asList() and add its elements using addAll().
6. Use contains() to check whether a particular student exists.
7. Use containsAll() to check whether all elements of another collection are present.
8. Display the number of elements using size() and check whether the collection is empty using isEmpty().
9. Remove one element using remove().
10. Remove all elements of another collection using removeAll().
11. Create an Iterator and traverse the remaining elements using hasNext() and next().

12. Remove all remaining elements using clear().
13. Use isEmpty() again to verify that the collection is empty.
14. Stop the program.

4. UML Class Diagram



The main class creates a Collection reference backed by an ArrayList. The program also uses Iterator for traversal and Arrays.asList() to create a collection of additional students.

5. Source Code

```
import java.util.ArrayList;
import java.util.Arrays;
import java.util.Collection;
import java.util.Iterator;

class CollectionMethodsDemo {

    public static void main(String[] args) {

        // Creating a Collection
        Collection<String> students = new ArrayList<>();

        // add(E e)
        students.add("Rahul");
        students.add("Priya");
        students.add("Arjun");

        System.out.println("After add(): " + students);

        // addAll(Collection c)
        Collection<String> newStudents =
            Arrays.asList("Anu", "Kiran");

        students.addAll(newStudents);

        System.out.println("After addAll(): " + students);

        // contains(Object o)
        System.out.println("Contains Rahul? "
            + students.contains("Rahul"));

        // containsAll(Collection c)
        System.out.println("Contains Anu and Kiran? "
            + students.containsAll(newStudents));

        // size()
        System.out.println("Size: " + students.size());

        // isEmpty()
        System.out.println("Is collection empty? "
            + students.isEmpty());

        // remove(Object o)
        students.remove("Arjun");

        System.out.println("After remove(): " + students);

        // removeAll(Collection c)
        students.removeAll(newStudents);

        System.out.println("After removeAll(): " + students);

        // iterator()
        System.out.println("Using iterator():");

        Iterator<String> iterator = students.iterator();

        while (iterator.hasNext()) {
            System.out.println(iterator.next());
        }

        // clear()
        students.clear();

        System.out.println("After clear(): " + students);

        // isEmpty() after clear
        System.out.println("Is collection empty now? "
            + students.isEmpty());
    }
}
```

6. Completion and Execution Commands

Step 1: Save the program as:
CollectionMethodsDemo.java

Step 2: Compile the program:
javac CollectionMethodsDemo.java

Step 3: Run the program:
java CollectionMethodsDemo

7. Output

```
After add(): [Rahul, Priya, Arjun]
After addAll(): [Rahul, Priya, Arjun, Anu, Kiran]
Contains Rahul? true
Contains Anu and Kiran? true
Size: 5
Is collection empty? false
After remove(): [Rahul, Priya, Anu, Kiran]
After removeAll(): [Rahul, Priya]
Using iterator():
Rahul
Priya
After clear(): []
Is collection empty now? true
```

Result: The Java program was successfully implemented and executed. The commonly used Collection interface methods were demonstrated using an ArrayList.